AI CLUSTERS



Scale Across:

To “Go Forward” or To “Go Backward”, That is the question

Jai Kumar

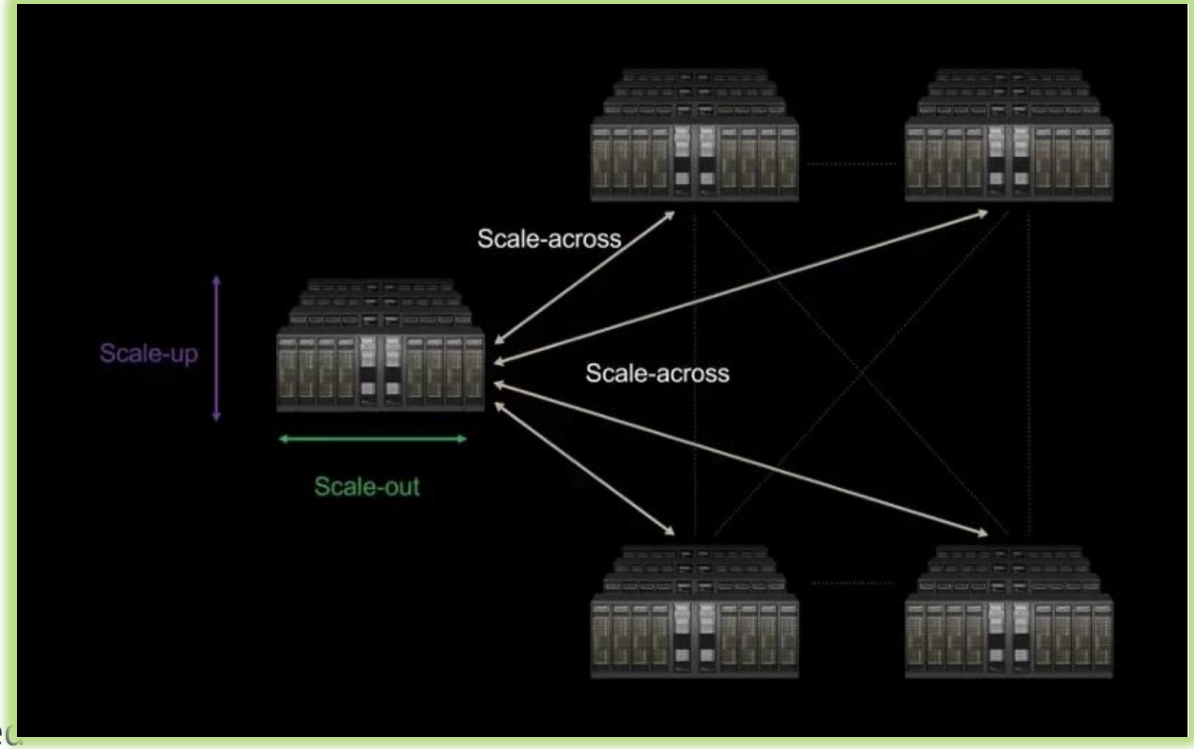
Distinguished Engineer, Broadcom Inc.

Tom Emmons

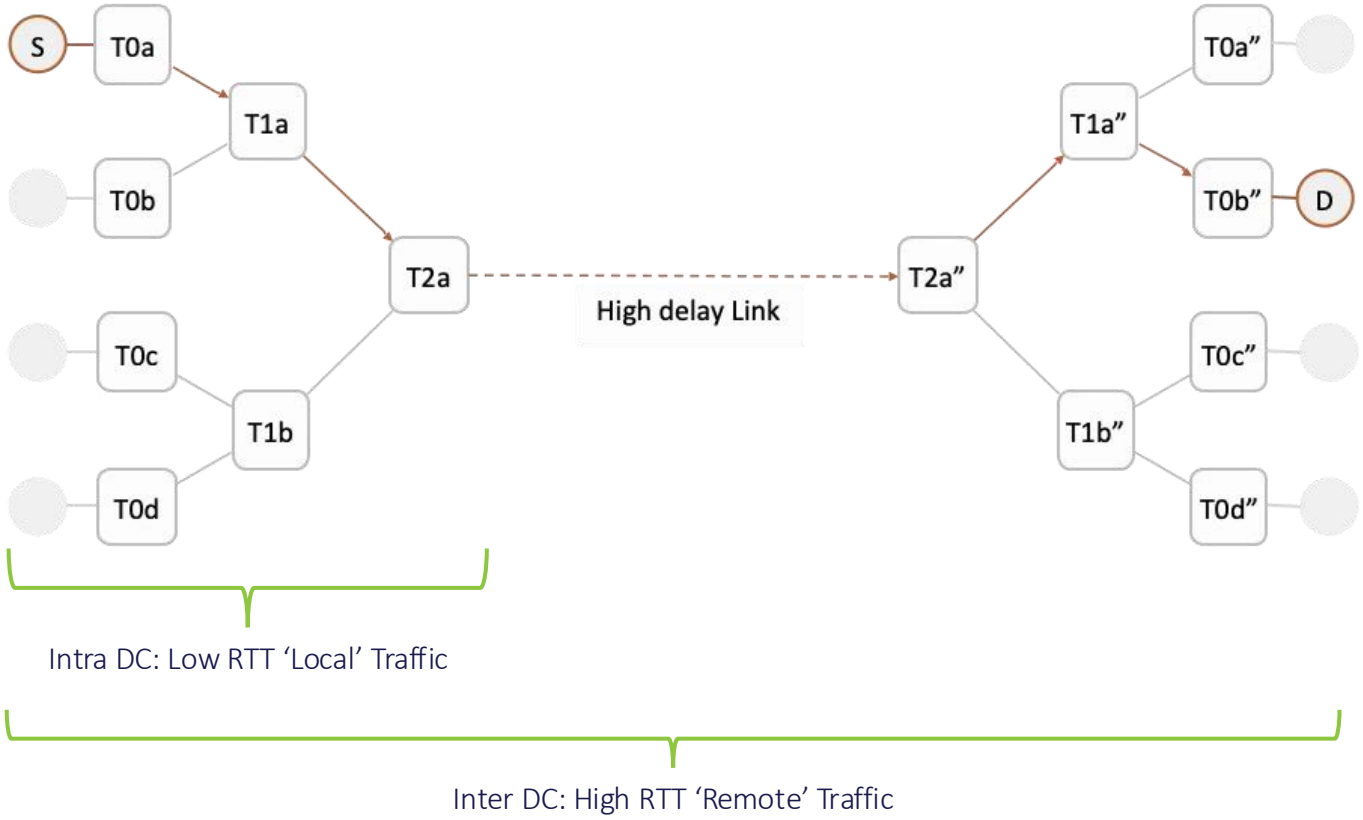
Software Engineer, Arista Networks Inc.

AI Fabric Domains

- Scale Up
  - Scope: Single Server Rack, Racks or Localized POD (< 30 meters)
  - Topology: Single Hop, High Radix switch connectivity, DAC
  - Focus: Low Latency, Massive Shared Memory Pool
- Scale Out
  - Scope: >1K GPUs across Racks
  - Topology: Multi Plane, Flat DC. Rail-optimized 2 or 3 Tier Leaf-Spine
  - Focus: Maximize Aggregate BW, Reliable Transport
- Scale Across
  - Scope: Physically separated data center campuses and regions (100+ km)
  - Topology: Routed Metro/Regional WAN with DCI
  - Focus: Avoiding timeout errors from packet loss. For synchronized flow arrivals, promptly detect packet loss despite long-RTT links



Reference Network Model



Challenges

- Incast like scenario at the T2a Gateway
- High latency for sender to learn of drops or congestion at the bottleneck queue before the high-delay link
- Characterization of congestion state and path failure are different in local and remote DC
- FCT and Tail Latency gets impacted adversely by long-delay links

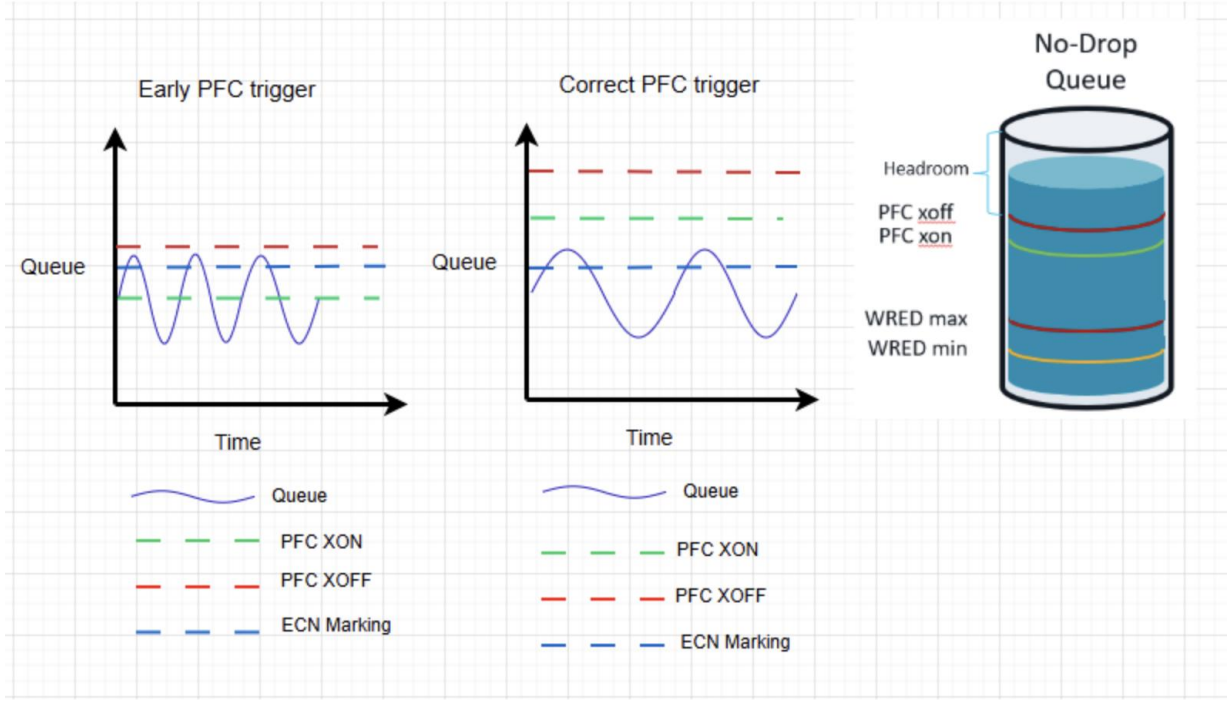
Existing Solutions and Shortcomings

1. TRIMMING:
   - Forward notification for queue congestion drop
   - No notification of incipient congestion or link failures

2. CSIG:
   - Forward notification for summary update
   - No notification of link failures

3. ECN
   - Proactive congestion signal
   - Single bit signal takes time to converge
   - Hard to tune

4. PFC:
   - Reactive link-level flow control. Stop Signal
   - Can cause pause storm.
   - Hard to tune



Distance – the fundamental issue

1. Long RTT slows feedback
   - Trimming, CSIG, and ECN all fundamentally require a RTT before the sender can react
   - Slow reaction times makes control loops more unstable
   - AI flows may be less than 1 RTT

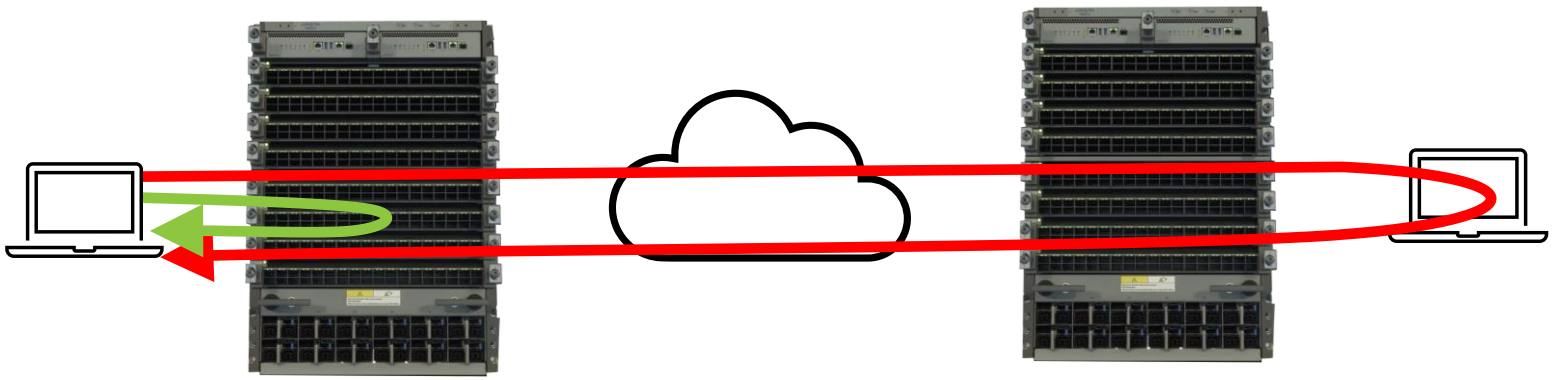


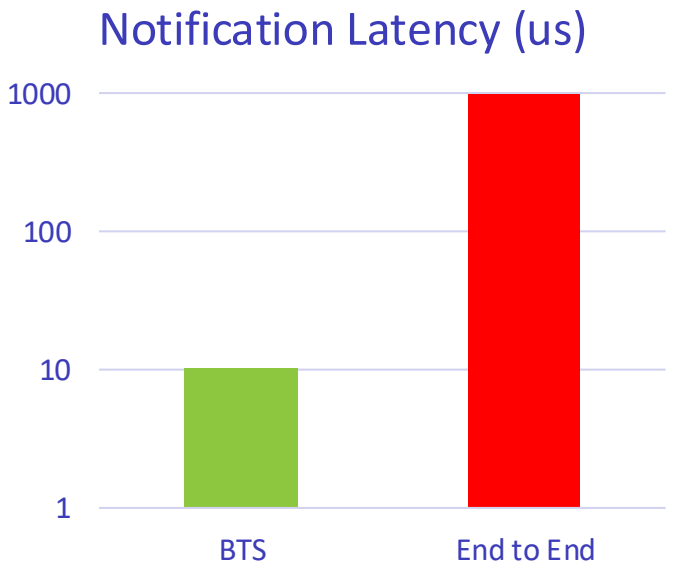
2. Long delay link is bandwidth constrained
   - Congestion notifications take critical bandwidth on expensive DCI links

3. Congestion Control Algorithms:
   - Today expect a homogenous fabric in terms of RTT delay and incast points
   - There are two heterogeneous RTT delay profiles in Scale Across
   - Incast is not just experienced at the last hop but is also experienced at the DCI link
   - Path Failures are not accounted for and usually are handled using timeouts

*image source - Gemini

Back To Sender





Requirements for End Points in Scale-X

Provide rich information to the Congestion Control Algorithm

1. Need to identify the adverse condition in the fabric
   - Queue congestion threshold breach
   - Queue congestion drop
   - Path failure

2. Associated Metadata
   - Port, Traffic Class
   - Device Information
   - Port utilization and Queue depth

3. Transport Context
   - Truncated packet

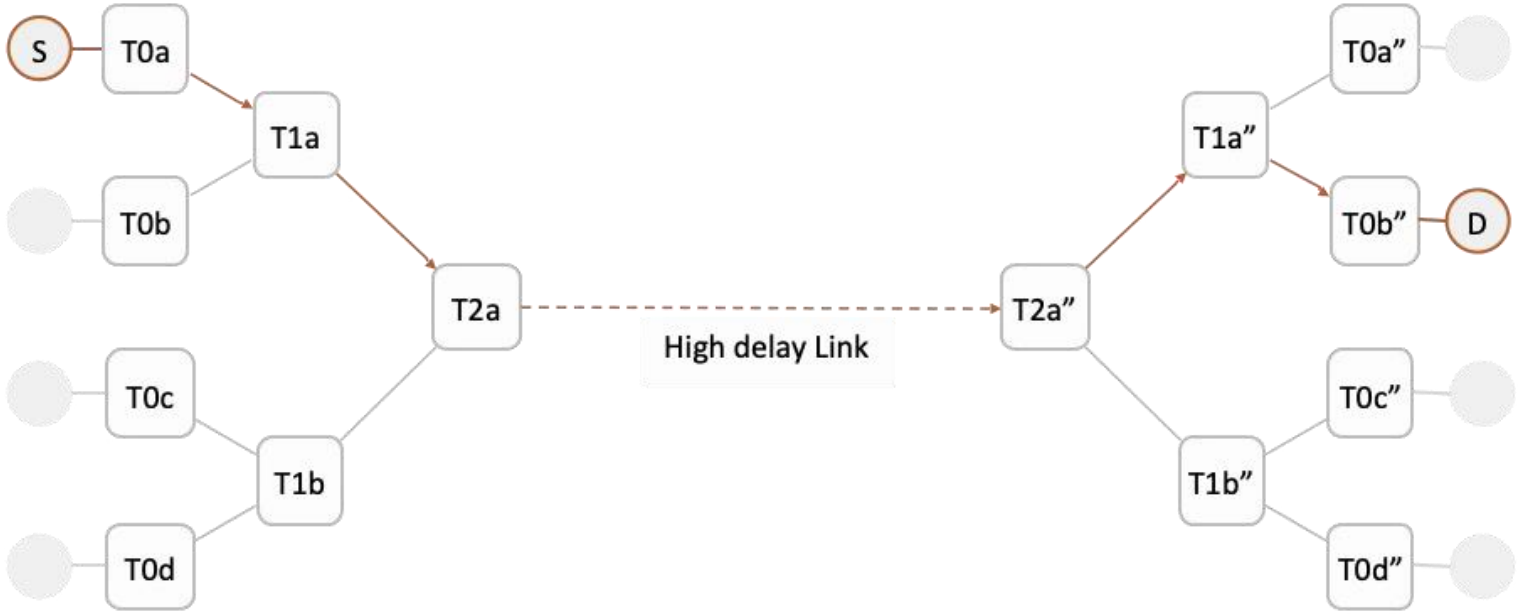
No impact to transport header

No impact to forwarding headers

No flow state maintenance at switch

~1% or less BW utilization by this control traffic

Questions to be Answered



- How to deal with congestion at multiple hops along the path?
- When a packet experiences congestion at one hop and is later dropped, should both hops notify the sender?
- How should the distance from sender factor into the notifications?
- Notifications are a new packet. How to rate control?
- How does the sender consume a high rate of notifications?
- Who performs the rate limited, sender or the switch generating the notifications?
- How does the CCA consume two disparate type of notifications i.e. DC local and DC remote?

Call to Action

- Join OCP SAI Workstream for SAI modeling
  - https://github.com/opencomputeproject/SAI/pull/2349
  - More work to be done and under discussion
- Join UEC BTS WG Adhoc
  - https://ultraethernet.org/

Thank You!

